

ORDER PAGE OVERHAUL - Complete this in order. Finish Part 1 completely before starting Part 2. Finish Part 2 completely before starting Part 3. And so on. Do not skip ahead.

=== PART 1: BUG FIXES (do these first, they are blocking real usage) ===

Fix the limit order submission error. The Schwab API returns "Limit price must be populated only for limit orders." This happens because the order is being sent as a market order type but with a limit price field populated. The fix is simple: every order submitted from this page is a limit order because the user always sets a price through the bid/mid/nat selector. Force the order type to LIMIT on every submission. Remove any conditional logic that tries to figure out whether it should be MARKET or LIMIT. It is always LIMIT. Also add a check: if the price field is empty or zero or undefined, block the submission and show an inline error message saying a price is required. Never let a priceless order reach the Schwab API.

Fix the NaN DTE display. The Pre-Trade Risk Check shows "NaN DTE - safe range" under DTE/Gamma Risk. The expiration date is not being parsed into a proper date object. Fix the parsing so DTE equals the number of calendar days between today and expiration. Use UTC midnight for both dates to avoid off-by-one timezone errors. Display it like "23 DTE - safe range" instead of "NaN DTE". If the date still cannot be parsed for some reason, show "DTE unavailable" instead of NaN.

Fix the IV display. The Pre-Trade Risk Check shows "Avg IV 6952%" which is obviously wrong. The raw value is a decimal like 0.6952 and it needs to be multiplied by 100 to become 69.52%. Format it to two decimal places with a percent sign. If the resulting percentage is above 500%, still show the number but add a small warning note next to it saying "unusually high - verify data" because that likely means bad data from a near-expiration or illiquid contract.

Finish all three bug fixes and confirm they work before moving to Part 2.

=== PART 2: STRATEGY NAME AND IDENTIFICATION ===

Add automatic strategy type detection so the order page displays the name of the strategy the user is trading. This name should appear prominently near the top of the page between the ticker header and the leg details. It should be clearly visible and immediately tell the user what they are doing without having to mentally parse the individual legs.

Detect the strategy by analyzing the legs:

If there are two legs, both puts, same expiration, and one is short at a higher

strike while the other is long at a lower strike, that is a Bull Put Spread. Display the name in green.

If there are two legs, both calls, same expiration, and one is short at a lower strike while the other is long at a higher strike, that is a Bear Call Spread. Display the name in red.

If there are two legs, both puts, same expiration, short at the lower strike and long at the higher strike, that is a Bear Put Spread (debit). Display in red.

If there are two legs, both calls, same expiration, short at the higher strike and long at the lower strike, that is a Bull Call Spread (debit). Display in green.

If there are four legs that form two vertical spreads (one put spread and one call spread) at the same expiration, that is an Iron Condor. Display in yellow or neutral color.

If there are four legs forming two vertical spreads at the same strikes but different types, that is an Iron Butterfly. Display in yellow or neutral.

If there are two legs at the same strike but different expirations and same type, that is a Calendar Spread. Display in neutral.

If there are two legs at different strikes and different expirations and same type, that is a Diagonal Spread. Display in neutral.

If there are three legs where two share the same strike, that is a Butterfly. Display in neutral.

A single long call is a Long Call in green. A single long put is a Long Put in red.

A single short call or short put with no protective leg is a Naked Position. Display in red with a warning. More on warnings in Part 3.

Anything else that does not match a known pattern is a Custom Multi-Leg. Display the number of legs and whether the net is a credit or debit.

Also add the strategy name to the confirmation modal that appears when the user taps Review Strategy Order. The confirmation modal should show the strategy name at the top, the word LIMIT ORDER as the order type, then the legs, then the net price and estimated credit or debit.

Finish strategy detection and confirm it correctly identifies at least a bull put spread, bear call spread, iron condor, and single leg before moving to Part 3.

=== PART 3: RISK WARNINGS ===

Add inline risk warnings that appear on the order page when certain conditions are detected. These are informational warnings, not blocking popups. The user can still proceed but the risks are clearly visible.

Naked position warning: If the strategy detection from Part 2 identifies a naked short call or naked short put (a short option with no protective long option), display a red warning banner on the order page that says: WARNING: This is a naked short position with undefined risk. Max loss is theoretically unlimited for calls and substantial for puts. Consider adding a protective leg to define your risk.

Wide bid-ask per-leg warning: The AI Co-Pilot section already warns about wide bid-ask spreads overall. Add a per-leg indicator: if any individual leg has a bid-ask spread wider than 200% of the mid price, put a small warning icon or highlight next to that specific leg's bid/ask numbers so the user can see which leg has the liquidity problem.

Earnings proximity warning: If the underlying stock has an earnings date that falls between today and the trade's expiration date, show a yellow warning: Earnings on [date] falls within this trade's expiration window. Expect elevated IV and potential large price moves. If earnings date data is not available yet (waiting on FMP integration), skip this warning for now.

Unusual size warning: If the order quantity is more than 20% of the average daily options volume on any leg's strike, show an orange warning: Order size is large relative to typical volume at this strike. Fill may be partial or delayed.

Insufficient Greeks warning: If delta or other Greeks are null or zero on any leg that should have meaningful values (not a deep ITM or far OTM edge case), show a gray notice: Greeks unavailable for one or more legs. Position-level risk metrics may be incomplete.

Outside trading hours notice: If the user tries to submit when the market is closed and extended hours is toggled off, show: Market is closed. This order will be queued for the next trading session. If extended hours is on, show: This order will be submitted for extended hours execution. Liquidity may be limited.

Finish all warnings and confirm they display correctly before moving to Part 4.

=== PART 4: VISUAL HIERARCHY AND PAGE STRUCTURE ===

The order page currently shows a lot of good information but everything has the same visual weight. Nothing stands out as more important than anything else. The user's eye does not know where to go first. Fix this by giving the page a clear top-to-bottom priority structure. The principle is: the most critical information (what am I trading, what does it cost, what is my max risk) should be the biggest and most prominent. Secondary information (Greeks, routing options, advanced

settings) should be present but visually quieter and not competing for the same attention.

Think about how thinkorswim structures its order page. The trade identity and key risk numbers are front and center. The details are accessible but not overwhelming. Apply that same thinking here.

At the top of the page: the ticker, price, and percent change (already there). Directly below that, the strategy name from Part 2 should be large and clearly visible. This whole top area should immediately answer the question "what am I about to trade" without scrolling or reading fine print.

Below the strategy name: the leg builder with the individual legs. This is already functional and does not need major changes.

Below the legs: the Strategy Preview card with Net Credit/Debit, Max Risk, Max Reward, R:R ratio, Breakeven, and Est. PoP. Make Net Credit or Net Debit and Max Risk the two most prominent numbers in this card. They should be slightly larger or bolder than the other metrics because they are the two numbers a trader looks at most while deciding on the trade. If technically feasible, make this section sticky so it stays visible when the user scrolls down to the price adjuster and quantity selector. If sticky causes problems on mobile, an alternative is a small floating bar at the top of the screen showing just Net Credit and Max Risk when the full card scrolls out of view.

Below the strategy preview: the position Greeks row showing Delta, Gamma, Theta, Vega. Change the display labels from Greek symbols to the actual words. Show "Delta" instead of the triangle symbol, "Gamma" instead of the gamma symbol, "Theta" instead of the theta symbol, "Vega" instead of the V symbol. The symbols look sleek but the words are clearer.

Below the Greeks: the price section with the Net Credit Price input, the BID/MID/NAT selector, and the bid-ask spread slider. Then the quantity selector. Remove the 50 and 100 preset buttons from the quantity row. Keep 1, 5, 10, 25. The plus/minus and manual input still allow any quantity but 50 and 100 suggest position sizes that are too large for most users.

Below quantity: the Exchange, Duration, and Extended Hours row. Collapse this into an expandable "Advanced Settings" section that is closed by default. BEST, DAY, and Off are the correct defaults 99% of the time so they do not need to take up visible space. The user can tap to expand if they need to change them.

Below advanced settings: the AI Co-Pilot section with the bid-ask warning and fill probability. Keep this exactly as it is.

Below AI Co-Pilot: the Pre-Trade Risk Check section. Make this collapsed by default showing just the pass or blocked badge with a chevron to expand. When

expanded it shows the full checklist of Pulse Alignment, R:R, Liquidity, Position Size, Vol Environment, DTE/Gamma Risk, and Leg Alignment. This is excellent institutional-grade functionality but it does not need to be fully expanded all the time.

At the bottom: the Review Strategy Order button. Keep as is.

Any risk warnings from Part 3 should appear inline near the relevant section. The naked position warning goes right below the strategy name since it is the most critical. Earnings and liquidity warnings go near the legs or the AI Co-Pilot section.

Finish the hierarchy restructuring and confirm the page reads clearly from top to bottom before moving to Part 5.

=== PART 5: REAL-TIME GREEKS AND METRICS UPDATING ===

When the user changes anything in the order builder, all the calculated values should update immediately without requiring a page refresh or a separate action.

When the user changes a strike on any leg: recalculate and update the strategy name, the Strategy Preview metrics (Net Credit, Max Risk, Max Reward, R:R, Breakeven, PoP), and the position Greeks using the new strike's data from the chain.

When the user changes quantity: scale all dollar values (Net Credit, Max Risk, Max Reward) and all Greeks proportionally.

When the user adjusts the price using the bid/mid/nat selector or the slider: update Net Credit/Debit, Max Risk, Max Reward, and R:R. Greeks do not change based on limit price.

When the user adds or removes a leg: reclassify the strategy type and recalculate all metrics and Greeks for the new leg combination.

These updates should feel instant for simple math like quantity scaling. If changing a strike requires fetching new Greeks from the Schwab API, show a brief inline loading indicator on just the values that are updating, not a full-page loading screen.

Finish real-time updating and confirm it works when changing strikes, quantity, and price before moving to Part 6.

=== PART 6: FIX EXISTING ROLL AND CLOSE POSITION FLOWS ===

The Roll Position and Close Position buttons already exist on the portfolio page but they are not working correctly. Roll Position does not populate the order

builder with the closing legs plus new opening legs. Verify and fix the following:

Close Position flow: When the user taps Close Position on an existing position from the portfolio page, the order builder should pre-populate with the inverse of the current position's legs. If the user is long a 405/400 bull put spread (short 405 put, long 400 put), closing means buying the 405 put and selling the 400 put. The legs should auto-populate with the correct directions and the same quantity as the existing position. The price should default to the mid price of the closing spread. Support partial closes by allowing the user to reduce the quantity to less than the full position size. When closing a partial position, display how many contracts will remain after the close.

Roll Position flow: When the user taps Roll Position on an existing position, the order builder should show two sections. First, the closing legs (same as Close Position, the inverse of the current position). Second, the opening legs for the new position, which should default to the same strategy type at the next available expiration with similar strikes. The user can then adjust the new expiration and strikes. Display the net credit or debit of the entire roll (the combined price of closing the old position and opening the new one). The confirmation modal should show all four legs (or however many) clearly labeled as closing legs and opening legs.

Verify that both flows correctly populate the order builder, calculate the combined metrics, and submit to Schwab as a single multi-leg order. If either flow is partially built but broken, fix what is broken rather than rebuilding from scratch. Test with a real or simulated position.

=== PART 7: STRATEGIST GREEKS INTEGRATION ===

This is a separate change from the order page. This updates the AI strategist to use per-strike Greeks (especially delta) when selecting strikes for trade recommendations.

Make sure the options chain data passed to the Claude prompt includes delta, gamma, theta, and vega for each contract. Schwab's options chain API returns these values per contract. Include them in the data that gets sent to Claude alongside strike, expiration, type, bid, ask, open interest, and volume. Make sure deltas are passed as numbers not strings. Puts should have negative deltas and calls positive. Pass fresh Greeks at the time of the recommendation, not cached values, because deltas shift as the underlying price moves throughout the day. If Greeks are null on a contract (common on far out of the money or illiquid strikes), pass null and tell the strategist to skip those strikes.

Update the Claude system prompt for the strategist to add these delta-based strike selection rules:

For credit spreads like bull put spreads and bear call spreads: target the short

strike at approximately 30 delta. If IV Rank is above 50, shift to 25 delta for higher probability. If the scanner gave very strong directional conviction, the short strike can move to 35 delta for more premium. For the long protective leg, use a spread width appropriate for the stock price: \$2.50 wide for stocks under \$100, \$5 wide for stocks \$100 to \$300, \$5 or \$10 wide for stocks above \$300.

For iron condors: target both short strikes at approximately 16 delta (one standard deviation). Long wings 1 to 2 strikes further out on each side.

For debit spreads: long leg at 45 to 50 delta (near the money), short leg 1 to 2 strikes further out.

For calendars: same strike on both legs targeting 50 delta (at the money).

For butterflies: center strikes at 50 delta with wings at equal distances.

If no strike has a delta within 0.05 of the target (happens on tickers with wide strike spacing), select the nearest available delta and mention it in the recommendation: "Nearest available delta to 30 target was 0.38 at the 395 strike. Strike spacing is wide on this name."

Prefer monthly expiration cycles over weeklies for credit spreads because of better liquidity and more stable Greeks. Use weeklies only when the target DTE falls between monthly cycles and the weekly has open interest above 100 contracts on the target strikes.

When the strategist presents a trade recommendation, include the delta of each leg in the output. Example: Sell the 405 put (delta -0.30) and buy the 400 put (delta -0.22) for a net credit of \$1.20. Also state why that delta was chosen: Selected 30-delta short strike for a balance of premium and probability. Or: Shifted to 25-delta short strike due to elevated IV Rank of 72 percent. After the individual legs, include the net position Greeks for the whole strategy.

Also display the delta next to each leg on the strategist card in the app so the user sees delta context without navigating to the chain.

Finish the strategist Greeks integration and confirm the strategist is now selecting strikes by delta target rather than by price proximity.